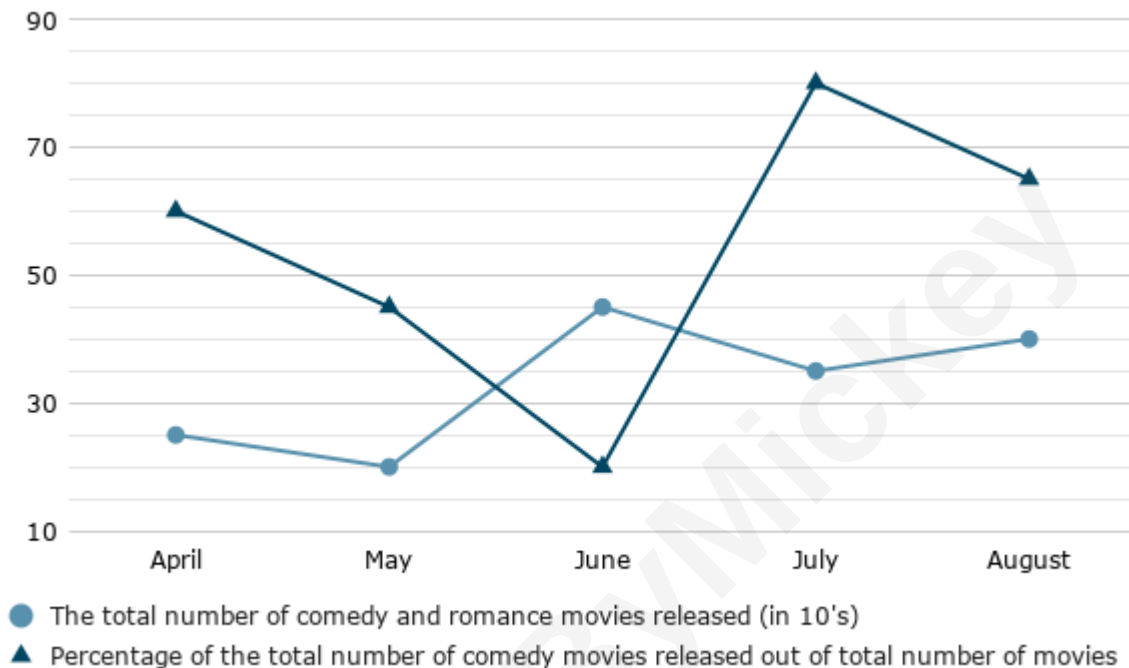


1. Questions

Study the following instructions carefully and answer the question given below.

The given line graph shows the total number of comedy and romance movies released (in 10's) in India across five different months, as well as the percentage of comedy movies out of the total releases during those same months.



The ratio of the total number of movies released in May that are hits to those that are flops is 3 : 2. If the total number of hit comedy movies released is 60, find the total number of flop romance movies released.

- a. 60
- b. 50
- c. 80
- d. 30
- e. 40

2. Questions

The average number of comedy movies released in July, August, and September is 290. Find the total number of comedy movies released in September and April.

- a. 320
- b. 630
- c. 360
- d. 520

e. 480

3. Questions

The total number of movies released in March is 2.5 times the total number of romance movies released in May. The total number of comedy movies released in March is 75 more than the total number of romance movies released in July. Find the total number of romance movies released in March.

- a. 170
- b. 150
- c. 130
- d. 140
- e. 160

4. Questions

If the total number of action movies released in June is 50% more than the sum of the total number of comedy movies released in July and August, find the total number of action movies released in June.

- a. 1020
- b. 720
- c. 930
- d. 810
- e. 680

5. Questions

Find the ratio of the total number of romance movies released in April and August to the total number of comedy movies released in April and May.

- a. 1 : 1
- b. 1 : 2
- c. 2 : 1
- d. 3 : 1
- e. 2 : 3

6. Questions

Study the following instructions carefully and answer the question given below.

The table below shows the total number of sweet boxes sold across four different shops, along with the percentage breakdown of laddu boxes, gulab jamun boxes, and rasgulla boxes sold by each shop.

Note: The total number of sweet boxes sold = (Total number of laddu boxes sold) + (Total number of gulab jamun boxes sold) + (Total number of rasgulla boxes sold).

Name of the shop	The total number of sweet boxes sold	Percentage of the total number of laddu boxes sold	Percentage of the total number of gulab jamun boxes sold	Percentage of the total number of rasgulla boxes sold
Shop A	480	50%	40%	—
Shop B	720	30%	—	20%
Shop C	—	—	—	60%
Shop D	400	—	25%	—

Note:

- i). The total number of laddu boxes sold in Shop A = The total number of sweet boxes sold in Shop C
- ii). The total number of laddu boxes sold in Shop D = 200 less than the total number of gulab jamun boxes sold in Shop B
- iii). The total number of laddu boxes sold in Shop C = The total number of gulab jamun boxes sold in Shop C

The total number of laddu boxes sold in Shop A is what percentage more than the total number of rasgulla boxes sold in Shop C?

- a. 25%
- b. 14.57%
- c. 66.67%
- d. 50%
- e. 33.33%

7. Questions

One laddu box contains 5 laddus, one gulab jamun box contains 10 gulab jamuns, and one rasgulla box contains 12 rasgullas. Find the total number of sweets sold in Shop D.

- a. 3160
- b. 4840
- c. 3820
- d. 4520
- e. 3480

8. Questions

The ratio of the total number of gulab jamun boxes manufactured in Shop B to the total number of rasgulla boxes manufactured in Shop D is 3 : 2. The ratio of the total number of gulab jamun boxes unsold in Shop B to the total number of rasgulla boxes unsold in Shop D is 1 : 1. Find the total number of gulab jamun boxes unsold in Shop B.

- a. 580
- b. 660
- c. 330
- d. 300
- e. 500

9. Questions

The ratio of the total number of rasgulla boxes sold in Shop E to the total number of rasgulla boxes sold in Shop A is 13 : 12. Find the total number of rasgulla boxes sold in Shop E and Shop B combined.

- a. 196
- b. 225
- c. 169
- d. 178
- e. 284

10. Questions

Find the difference between the total number of laddu boxes sold in Shop B and Shop C combined, and the total number of gulab jamun boxes sold in Shop A and Shop C combined.

- a. 48
- b. 60
- c. 24
- d. 36
- e. 12

11. Questions

Rajesh spends 20% of his monthly salary on groceries. From the remaining salary, 12.5% is spent on buying books, and 7% is spent on rent. He saves the rest of his income. If the amount he spends on rent is Rs. 5,600, find his annual savings.

- a. Rs. 772800
- b. Rs. 682800

- c. Rs. 782400
- d. Rs. 678400
- e. Rs. 723800

12. Questions

An article is marked up by 30% above its cost price. The shopkeeper offers a discount of Rs. 320. If the final selling price yields a profit of 20% over the cost price, find the selling price of the article required to make a profit of 45%.

- a. Rs. 3680
- b. Rs. 4480
- c. Rs. 3260
- d. Rs. 4640
- e. Rs. 5600

13. Questions

The ratio of the speed of a boat in still water to the speed of the stream is 13:5. The boat travels (D - 30) km upstream and (D + 40) km downstream in 3 hours and 45 minutes. If the speed of the stream is 10 km/h, find the value of D.

- a. 60
- b. 70
- c. 30
- d. 50
- e. 40

14. Questions

A box contains three types of rings: gold, silver, and diamond. There are 5 gold rings, 10 silver rings, and 3 diamond rings. If three rings are selected at random, find the probability of getting exactly one gold ring, one silver ring, and one diamond ring.

- a. 25 / 136
- b. 21 / 136
- c. 27 / 136
- d. 29 / 136
- e. 31 / 136

15. Questions

The total amount received after investing a certain sum of money at a simple interest rate of 7% per annum for 8 years is Rs. 7800. If the same sum of money is invested at a compound interest rate of 10% per annum for 2 years, find the total amount received.

- a. Rs. 6080
- b. Rs. 7080
- c. Rs. 6050
- d. Rs. 6500
- e. Rs. 7500

16. Questions

The average of the present ages of a Dad, Mom, and son is 45 years. The age of the Dad, 5 years ago was the same as the present age of the Mom. The ratio of the ages of the Mom and son 10 years ago was 2:1. Find the age of the dad five years from now.

- a. 65 years
- b. 50 years
- c. 55 years
- d. 60 years
- e. 35 years

17. Questions

A vessel contains a mixture of juice and water in the ratio of 4 : 5. If the initial quantity of juice is increased by 25% and 30 litres of water is added to the mixture, the ratio of juice to water becomes 1 : 2. Find the initial quantity of mixture in the vessel.

- a. 64 litres
- b. 72 litres
- c. 54 litres
- d. 68 litres
- e. 45 litres

18. Questions

12 men can complete 20% of a work in 5 days, while 15 women can complete 50% of the same work in 10 days. If 5 men and 10 women work together, how many days will they take to complete 120% of the total work?

- a. 48 days
- b. 60 days

- c. 12 days
- d. 36 days
- e. 24 days

19. Questions

A and B invested ₹650 and ₹550 respectively. After a (x) number of months, A increased his investment by $(1/13)^{\text{th}}$ and B decreased his investment by $(1/11)^{\text{th}}$. At the end of the year, the ratio of the profit share of B to A is 4 : 5, find the value of x.

- a. 8
- b. 9
- c. 7
- d. 6
- e. 5

20. Questions

The area of a circle is the same as the base area of a cylinder. If the perimeter of the circle is 88 cm and the curved surface area of the cylinder is 1848 cm^2 , find the height of the cylinder.

- a. 35 cm
- b. 21 cm
- c. 7 cm
- d. 14 cm
- e. 28 cm

21. Questions

What value should come in the place of (?) in the following questions?

98% of 300 + (720 / 12) = (?) * 3

- a. 118
- b. 102
- c. 124
- d. 108
- e. 112

22. Questions

$\sqrt{841} + (145 / 5) - (8 * 6) = (?)$

- a. 30
- b. 25
- c. 15
- d. 10
- e. 20

23. Questions

$$(12 * 16 * 32) / (8 * 2 * 6) = \sqrt[3]{(?) * 4}$$

- a. 284
- b. 400
- c. 256
- d. 196
- e. 169

24. Questions

$$(98 / 2) - (143 / 11) = (?) - \sqrt{49} + 19$$

- a. 60
- b. 24
- c. 36
- d. 12
- e. 48

25. Questions

$$(12 * 15) + (7 * 6) - (9 * 3) = (?) + 35$$

- a. 120
- b. 140
- c. 150
- d. 180
- e. 160

26. Questions

What approximate value should come in the place of (?) in the following question?

$$23.90 + (123.88 / 4.09) = (?)^2 - (13.15 * 1.89)$$

- a. 11
- b. 9
- c. 8
- d. 12
- e. 7

27. Questions

$$45.12\% \text{ of } 579.91 - (155.94 / 13.04) = (?) * 3.01$$

- a. 83
- b. 79
- c. 95
- d. 89
- e. 76

28. Questions

$$\sqrt{(12.90 + 23.13)} = (?) - (14.11 * 14.89) + (2.07)^4$$

- a. 100
- b. 500
- c. 300
- d. 200
- e. 400

29. Questions

$$(5.11 * 8.91) - \sqrt{35} + (11.21 * 11.92) = (?)$$

- a. 177
- b. 171
- c. 163
- d. 185
- e. 167

30. Questions

$$(189.19 / 8.88) + \sqrt{(91.21 - 65.92)} = (?)^3 - 38.12$$

- a. 3

- b. 2
- c. 6
- d. 4
- e. 5

31. Questions

Find out the missing number in the following number series.

9, 28, 126, ?, 1332, 2198

- a. 65
- b. 344
- c. 217
- d. 730
- e. 513

32. Questions

123, ?, 292, 398, 521, 663, 826

- a. 201
- b. 196
- c. 189
- d. 213
- e. 222

33. Questions

71, 77, 89, ?, 161, 257, 449

- a. 119
- b. 113
- c. 115
- d. 104
- e. 107

34. Questions

35.5, 156.5, 300.5, 469.5, 665.5, ?, 1146.5

- a. 875.5

- b. 895.5
- c. 885.5
- d. 860.5
- e. 890.5

35. Questions**26.25, 52.5, ?, 70, 14, 84, 12**

- a. 30
- b. 17.5
- c. 20.5
- d. 25
- e. 35

36. Questions**Find out the wrong number in the following number series.****259, 220, 175, 128, 75, 16**

- a. 259
- b. 16
- c. 75
- d. 128
- e. 220

37. Questions**7560, 1680, 420, 120, 60, 16, 8**

- a. 420
- b. 60
- c. 120
- d. 16
- e. 8

38. Questions**11, 24, 88, 324, 1640, 9870**

- a. 9870

- b. 11
- c. 88
- d. 324
- e. 24

39. Questions

134, 142, 158, 182, 214, 264, 302

- a. 142
- b. 134
- c. 264
- d. 302
- e. 214

40. Questions

186, 170, 195, 165, 208, 144, 225

- a. 225
- b. 144
- c. 165
- d. 195
- e. 208

41. Questions

The following question contains two equations, I and II. You have to solve both equations and determine the relationship between them and give an answer.

I). $3x^2 - 41x + 136 = 0$

II). $4y^2 - 68y + 288 = 0$

- a. $x \leq y$
- b. $x = y$ or relationship cannot be established
- c. $x \geq y$
- d. $x < y$
- e. $x > y$

42. Questions

I). $4x + y = 15$

II). $5x + 6y = 14$

- a. $x = y$ or relationship cannot be established
- b. $x > y$
- c. $x < y$
- d. $x \leq y$
- e. $x \geq y$

43. Questions

I). $x^2 - 6x - 135 = 0$

II). $y^2 - 32y + 87 = 0$

- a. $x > y$
- b. $x \leq y$
- c. $x < y$
- d. $x = y$ or relationship cannot be established
- e. $x \geq y$

44. Questions

I). $x^2 + 15x - 450 = 0$

II). $y^2 + 34y + 288 = 0$

- a. $x > y$
- b. $x < y$
- c. $x \leq y$
- d. $x \geq y$
- e. $x = y$ or relationship cannot be established

45. Questions

I). $x^2 = \sqrt{49} + 29$

II). $y^2 - 17y + 72 = 0$

- a. $x \leq y$
- b. $x = y$ or relationship cannot be established

c. $x > y$

d. $x \geq y$

e. $x < y$

Explanations:

1. Questions

April:

The total number of comedy and romance movies released = 250

The total number of comedy movies released = 60% of 250

$$= (60 / 100) * 250 = 150$$

The total number of romance movies released = $250 - 150 = 100$

May:

The total number of comedy and romance movies released = 200

The total number of comedy movies released = 45% of 200

$$= (45 / 100) * 200 = 90$$

The total number of romance movies released = $200 - 90 = 110$

June:

The total number of comedy and romance movies released = 450

The total number of comedy movies released = 20% of 450

$$= (20 / 100) * 450 = 90$$

The total number of romance movies released = $450 - 90 = 360$

July:

The total number of comedy and romance movies released = 350

The total number of comedy movies released = 80% of 350

$$= (80 / 100) * 350 = 280$$

The total number of romance movies released = $350 - 280 = 70$

August:

The total number of comedy and romance movies released = 400

The total number of comedy movies released = 65% of 400

$$= (65 / 100) * 400 = 260$$

The total number of romance movies released = $400 - 260 = 140$

Name of the month	The total number of comedy and romance movies released	The total number of comedy movies released	The total number of romance movies released
April	250	150	100
May	200	90	110
June	450	90	360
July	350	280	70
August	400	260	140

Answer: B

The total number of movies released in May = 200

The total number of movies released in May that are hit : The total number of movies released in May that are flop = 3 : 2

The total number of movies released in May that are hit = $3x$

The total number of movies released in May that are flop = $2x$

$$3x + 2x = 200$$

$$5x = 200$$

$$x = 40$$

The total number of movies released in May that are hit = $3x = 3 * 40 = 120$

The total number of movies released in May that are flop = $2x = 2 * 40 = 80$

The total number of comedy movies released in May that are hit = 60

The total number of of comedy movies released in May that are flop = $90 - 60 = 30$

The total number of romance movies released in May that are flop = $80 - 30 = 50$

Hence, the correct answer is option B

2. Questions

April:

The total number of comedy and romance movies released = 250

The total number of comedy movies released = 60% of 250

$$= (60 / 100) * 250 = 150$$

The total number of romance movies released = $250 - 150 = 100$

May:

The total number of comedy and romance movies released = 200

The total number of comedy movies released = 45% of 200

$$= (45 / 100) * 200 = 90$$

The total number of romance movies released = $200 - 90 = 110$

June:

The total number of comedy and romance movies released = 450

The total number of comedy movies released = 20% of 450

$$= (20 / 100) * 450 = 90$$

The total number of romance movies released = $450 - 90 = 360$

July:

The total number of comedy and romance movies released = 350

The total number of comedy movies released = 80% of 350

$$= (80 / 100) * 350 = 280$$

The total number of romance movies released = $350 - 280 = 70$

August:

The total number of comedy and romance movies released = 400

The total number of comedy movies released = 65% of 400

$$= (65 / 100) * 400 = 260$$

The total number of romance movies released = $400 - 260 = 140$

Name of the month	The total number of comedy and romance movies released	The total number of comedy movies released	The total number of romance movies released
April	250	150	100
May	200	90	110
June	450	90	360
July	350	280	70
August	400	260	140

Answer: E

The total number of comedy movies released in April = 150

The total number of comedy movies released in July = 280

The total number of comedy movies released in August = 260

The average number of comedy movies released in July, August and September = 290

The total number of comedy movies released in July, August and September = 870

The total number of comedy movies released in September = $870 - 280 - 260 = 330$

Required answer = $330 + 150 = 480$

Hence, the correct answer is option E

3. Questions

April:

The total number of comedy and romance movies released = 250

The total number of comedy movies released = 60% of 250

$= (60 / 100) * 250 = 150$

The total number of romance movies released = $250 - 150 = 100$

May:

The total number of comedy and romance movies released = 200

The total number of comedy movies released = 45% of 200

$= (45 / 100) * 200 = 90$

The total number of romance movies released = $200 - 90 = 110$

June:

The total number of comedy and romance movies released = 450

The total number of comedy movies released = 20% of 450

$$= (20 / 100) * 450 = 90$$

The total number of romance movies released = $450 - 90 = 360$

July:

The total number of comedy and romance movies released = 350

The total number of comedy movies released = 80% of 350

$$= (80 / 100) * 350 = 280$$

The total number of romance movies released = $350 - 280 = 70$

August:

The total number of comedy and romance movies released = 400

The total number of comedy movies released = 65% of 400

$$= (65 / 100) * 400 = 260$$

The total number of romance movies released = $400 - 260 = 140$

Name of the month	The total number of comedy and romance movies released	The total number of comedy movies released	The total number of romance movies released
April	250	150	100
May	200	90	110
June	450	90	360
July	350	280	70
August	400	260	140

Answer: C

The total number of romance movies released in May = 110

The total number of romance movies released in July = 70

The total number of movies released in March = $2.5 * 110 = 275$

The total number of comedy movies released in March = $75 + 70 = 145$

The total number of romance movies released in March = $275 - 145 = 130$

Hence, the correct answer is option C

4. Questions

April:

The total number of comedy and romance movies released = 250

The total number of comedy movies released = 60% of 250

$$= (60 / 100) * 250 = 150$$

The total number of romance movies released = $250 - 150 = 100$

May:

The total number of comedy and romance movies released = 200

The total number of comedy movies released = 45% of 200

$$= (45 / 100) * 200 = 90$$

The total number of romance movies released = $200 - 90 = 110$

June:

The total number of comedy and romance movies released = 450

The total number of comedy movies released = 20% of 450

$$= (20 / 100) * 450 = 90$$

The total number of romance movies released = $450 - 90 = 360$

July:

The total number of comedy and romance movies released = 350

The total number of comedy movies released = 80% of 350

$$= (80 / 100) * 350 = 280$$

The total number of romance movies released = $350 - 280 = 70$

August:

The total number of comedy and romance movies released = 400

The total number of comedy movies released = 65% of 400

$$= (65 / 100) * 400 = 260$$

The total number of romance movies released = $400 - 260 = 140$

Name of the month	The total number of comedy and romance movies released	The total number of comedy movies released	The total number of romance movies released
April	250	150	100
May	200	90	110
June	450	90	360
July	350	280	70
August	400	260	140

Answer: D

The total number of **comedy** movies released in July = 280

The total number of **comedy** movies released in August = 260

The total number of action movies released in June = 150% of (280 + 260)

$$= (150 / 100) * 540$$

$$= 810$$

Hence, the correct answer is option D

5. Questions

April:

The total number of comedy and romance movies released = 250

The total number of comedy movies released = 60% of 250

$$= (60 / 100) * 250 = 150$$

The total number of romance movies released = 250 - 150 = 100

May:

The total number of comedy and romance movies released = 200

The total number of comedy movies released = 45% of 200

$$= (45 / 100) * 200 = 90$$

The total number of romance movies released = 200 - 90 = 110

June:

The total number of comedy and romance movies released = 450

The total number of comedy movies released = 20% of 450

$$= (20 / 100) * 450 = 90$$

The total number of romance movies released = 450 - 90 = 360

July:

The total number of comedy and romance movies released = 350

The total number of comedy movies released = 80% of 350

$$= (80 / 100) * 350 = 280$$

The total number of romance movies released = 350 - 280 = 70

August:

The total number of comedy and romance movies released = 400

The total number of comedy movies released = 65% of 400

$$= (65 / 100) * 400 = 260$$

The total number of romance movies released = 400 - 260 = 140

Name of the month	The total number of comedy and romance movies released	The total number of comedy movies released	The total number of romance movies released
April	250	150	100
May	200	90	110
June	450	90	360
July	350	280	70
August	400	260	140

Answer: A

The total number of romance movies released in April = 100

The total number of romance movies released in August = 140

The total number of comedy movies released in April = 150

The total number of comedy movies released in May = 90

Required answer = (100 + 140) : (150 + 90)

$$= 240 : 240 = 1 : 1$$

Hence, the correct answer is option A

6. Questions

Shop A:

The total number of sweet boxes sold = 480

The total number of laddu boxes sold = 50% of 480 = $(50 / 100) * 480 = 240$

The total number of gulab jamun boxes sold = 40% of 480 = $(40 / 100) * 480 = 192$

The total number of rasgulla boxes sold = $(100 - 90)\%$ of 480 = 10% of 480
 $= (10 / 100) * 480 = 48$

Shop B:

The total number of sweet boxes sold = 720

The total number of laddu boxes sold = 30% of 720 = $(30 / 100) * 720 = 216$

The total number of rasgulla boxes sold = 20% of 720 = $(20 / 100) * 720 = 144$

The total number of gulab jamun boxes sold = $(100 - 50)\%$ of 720 = 50% of 720 = $(50 / 100) * 720 = 360$

Shop C:

The total number of sweet boxes sold = The total number of laddu boxes sold in Shop A

The total number of sweet boxes sold = 240

The total number of rasgulla boxes sold = 60% of 240 = $(60 / 100) * 240 =$
144

The total number of laddu boxes sold = The total number of gulab jamun boxes sold

Let the total number of laddu boxes sold be x

The total number of sweet boxes sold = (Total number of laddu boxes sold) + (Total number of gulab jamun boxes sold) + (Total number of rasgulla boxes sold)

$$240 = x + x + 144$$

$$2x = 96$$

$$x = 48$$

The total number of laddu boxes sold = 48

The total number of gulab jamun boxes sold = 48

Shop D:

The total number of sweet boxes sold = 400

The total number of gulab jamun boxes sold = 25% of 400 = 100

The total number of laddu boxes sold = The total number of gulab jamun boxes sold in Shop B - 200

The total number of laddu boxes sold = $360 - 200 = 160$

The total number of rasgulla boxes sold = $400 - 100 - 160 = 140$

Name of the shop	The total number of sweet boxes sold	The total number of laddu boxes sold	The total number of gulab jamun boxes sold	The total number of rasgulla boxes sold
Shop A	480	240	192	48
Shop B	720	216	360	144
Shop C	240	48	48	144
Shop D	400	160	100	140

Answer: C

The total number of laddu boxes sold in Shop A = 240

The total number of rasgulla boxes sold in Shop C = 144

Required percentage = $((240 - 144) / 144) * 100$

= $(96 / 144) * 100 = 66.67\%$

Hence, the correct answer is option C

7. Questions

Shop A:

The total number of sweet boxes sold = 480

The total number of laddu boxes sold = 50% of 480 = $(50 / 100) * 480 = 240$

The total number of gulab jamun boxes sold = 40% of 480 = $(40 / 100) * 480 = 192$

The total number of rasgulla boxes sold = $(100 - 90)\%$ of 480 = 10% of 480

= $(10 / 100) * 480 = 48$

Shop B:

The total number of sweet boxes sold = 720

The total number of laddu boxes sold = 30% of 720 = $(30 / 100) * 720 = 216$

The total number of rasgulla boxes sold = 20% of 720 = $(20 / 100) * 720 = 144$

The total number of gulab jamun boxes sold = $(100 - 50)\%$ of 720 = 50% of 720 = $(50 / 100) * 720 = 360$

Shop C:

The total number of sweet boxes sold = The total number of laddu boxes sold in Shop A

The total number of sweet boxes sold = 240

The total number of rasgulla boxes sold = 60% of 240 = $(60 / 100) * 240 = 144$

The total number of laddu boxes sold = The total number of gulab jamun boxes sold

Let the total number of laddu boxes sold be x

The total number of sweet boxes sold = (Total number of laddu boxes sold) + (Total number of gulab jamun boxes sold) + (Total number of rasgulla boxes sold)

$$240 = x + x + 144$$

$$2x = 96$$

$$x = 48$$

The total number of laddu boxes sold = 48

The total number of gulab jamun boxes sold = 48

Shop D:

The total number of sweet boxes sold = 400

The total number of gulab jamun boxes sold = 25% of 400 = 100

The total number of laddu boxes sold = The total number of gulab jamun boxes sold in Shop B - 200

The total number of laddu boxes sold = $360 - 200 = 160$

The total number of rasgulla boxes sold = $400 - 100 - 160 = 140$

Name of the shop	The total number of sweet boxes sold	The total number of laddu boxes sold	The total number of gulab jamun boxes sold	The total number of rasgulla boxes sold
Shop A	480	240	192	48
Shop B	720	216	360	144
Shop C	240	48	48	144
Shop D	400	160	100	140

Answer: E

The total number of laddu boxes sold in Shop D = 160

The total number of gulab jamun boxes sold in Shop D = 100

The total number of rasgulla boxes sold in Shop D = 140

The total number of sweets sold in Shop D = $(160 * 5) + (100 * 10) + (140 * 12)$
 $= 800 + 1000 + 1680 = 3480$

Hence, the correct answer is option E

8. Questions

Shop A:

The total number of sweet boxes sold = 480

The total number of laddu boxes sold = 50% of 480 = $(50 / 100) * 480 = 240$

The total number of gulab jamun boxes sold = 40% of 480 = $(40 / 100) * 480 = 192$

The total number of rasgulla boxes sold = $(100 - 90)\%$ of 480 = 10% of 480
 $= (10 / 100) * 480 = 48$

Shop B:

The total number of sweet boxes sold = 720

The total number of laddu boxes sold = 30% of 720 = $(30 / 100) * 720 = 216$

The total number of rasgulla boxes sold = 20% of 720 = $(20 / 100) * 720 = 144$

The total number of gulab jamun boxes sold = $(100 - 50)\%$ of 720 = 50% of 720 = $(50 / 100) * 720 = 360$

Shop C:

The total number of sweet boxes sold = The total number of laddu boxes sold in Shop A

The total number of sweet boxes sold = 240

The total number of rasgulla boxes sold = 60% of 240 = $(60 / 100) * 240 =$
144

The total number of laddu boxes sold = The total number of gulab jamun boxes sold

Let the total number of laddu boxes sold be x

The total number of sweet boxes sold = (Total number of laddu boxes sold) + (Total number of gulab jamun boxes sold) + (Total number of rasgulla boxes sold)

$$240 = x + x + 144$$

$$2x = 96$$

$$x = 48$$

The total number of laddu boxes sold = 48

The total number of gulab jamun boxes sold = 48

Shop D:

The total number of sweet boxes sold = 400

The total number of gulab jamun boxes sold = 25% of 400 = 100

The total number of laddu boxes sold = The total number of gulab jamun boxes sold in Shop B - 200

The total number of laddu boxes sold = 360 - 200 = 160

The total number of rasgulla boxes sold = 400 - 100 - 160 = 140

Name of the shop	The total number of sweet boxes sold	The total number of laddu boxes sold	The total number of gulab jamun boxes sold	The total number of rasgulla boxes sold
Shop A	480	240	192	48
Shop B	720	216	360	144
Shop C	240	48	48	144
Shop D	400	160	100	140

Answer: D

The total number of gulab jamun boxes sold in Shop B = 360

The total number of rasgulla boxes sold in Shop D = 140

The total number of gulab jamun boxes manufactured in Shop B : The total number of rasgulla boxes manufactured in Shop D = 3 : 2

The total number of gulab jamun boxes manufactured in Shop B = 3x

The total number of rasgulla boxes manufactured in Shop D = 2x

The total number of gulab jamun boxes unsold in Shop B : The total number of rasgulla boxes unsold in Shop D = 1 : 1

The total number of gulab jamun boxes unsold in Shop B = y

The total number of rasgulla boxes unsold in Shop D = y

$$3x - y = 360 \rightarrow (1)$$

$$2x - y = 140$$

$$2x - 140 = y \rightarrow (2)$$

Substitute equation (2) in (1),

$$3x - 2x + 140 = 360$$

$$x = 220$$

Substitute $x = 220$ in equation (2),

$$(2 * 220) - 140 = y$$

$$440 - 140 = y$$

$$y = 300$$

The total number of gulab jamun boxes unsold in Shop B = $y = 300$

Hence, the correct answer is option D

9. Questions

Shop A:

The total number of sweet boxes sold = 480

The total number of laddu boxes sold = 50% of 480 = $(50 / 100) * 480 = 240$

The total number of gulab jamun boxes sold = 40% of 480 = $(40 / 100) * 480 = 192$

The total number of rasgulla boxes sold = $(100 - 90)\%$ of 480 = 10% of 480

$$= (10 / 100) * 480 = 48$$

Shop B:

The total number of sweet boxes sold = 720

The total number of laddu boxes sold = 30% of 720 = $(30 / 100) * 720 = 216$

The total number of rasgulla boxes sold = 20% of 720 = $(20 / 100) * 720 = 144$

The total number of gulab jamun boxes sold = $(100 - 50)\%$ of 720 = 50% of 720 = $(50 / 100) * 720 = 360$

Shop C:

The total number of sweet boxes sold = The total number of laddu boxes sold in Shop A

The total number of sweet boxes sold = 240

The total number of rasgulla boxes sold = 60% of 240 = $(60 / 100) * 240 =$

$$144$$

The total number of laddu boxes sold = The total number of gulab jamun boxes sold

Let the total number of laddu boxes sold be x

The total number of sweet boxes sold = (Total number of laddu boxes sold) + (Total number of gulab jamun boxes sold) + (Total number of rasgulla boxes sold)

$$240 = x + x + 144$$

$$2x = 96$$

$$x = 48$$

The total number of laddu boxes sold = 48

The total number of gulab jamun boxes sold = 48

Shop D:

The total number of sweet boxes sold = 400

The total number of gulab jamun boxes sold = 25% of 400 = 100

The total number of laddu boxes sold = The total number of gulab jamun boxes sold in Shop B - 200

The total number of laddu boxes sold = 360 - 200 = 160

The total number of rasgulla boxes sold = 400 - 100 - 160 = 140

Name of the shop	The total number of sweet boxes sold	The total number of laddu boxes sold	The total number of gulab jamun boxes sold	The total number of rasgulla boxes sold
Shop A	480	240	192	48
Shop B	720	216	360	144
Shop C	240	48	48	144
Shop D	400	160	100	140

Answer: A

The total number of rasgulla boxes sold in Shop A = 48

The total number of rasgulla boxes sold in Shop B = 144

The total number of rasgulla boxes sold in Shop E = $(13 / 12) * 48$

= 52

Required answer = 144 + 52 = 196

Hence, the correct answer is option A

10. Questions

Shop A:

The total number of sweet boxes sold = 480

The total number of laddu boxes sold = 50% of 480 = $(50 / 100) * 480 = 240$

The total number of gulab jamun boxes sold = 40% of 480 = $(40 / 100) * 480 = 192$

The total number of rasgulla boxes sold = $(100 - 90)\%$ of 480 = 10% of 480

$$= (10 / 100) * 480 = 48$$

Shop B:

The total number of sweet boxes sold = 720

The total number of laddu boxes sold = 30% of 720 = $(30 / 100) * 720 = 216$

The total number of rasgulla boxes sold = 20% of 720 = $(20 / 100) * 720 = 144$

The total number of gulab jamun boxes sold = $(100 - 50)\%$ of 720 = 50% of 720 = $(50 / 100) * 720 = 360$

Shop C:

The total number of sweet boxes sold = The total number of laddu boxes sold in Shop A

The total number of sweet boxes sold = 240

The total number of rasgulla boxes sold = 60% of 240 = $(60 / 100) * 240 =$
144

The total number of laddu boxes sold = The total number of gulab jamun boxes sold

Let the total number of laddu boxes sold be x

The total number of sweet boxes sold = (Total number of laddu boxes sold) + (Total number of gulab jamun boxes sold) + (Total number of rasgulla boxes sold)

$$240 = x + x + 144$$

$$2x = 96$$

$$x = 48$$

The total number of laddu boxes sold = 48

The total number of gulab jamun boxes sold = 48

Shop D:

The total number of sweet boxes sold = 400

The total number of gulab jamun boxes sold = 25% of 400 = 100

The total number of laddu boxes sold = The total number of gulab jamun boxes sold in Shop B - 200

The total number of laddu boxes sold = $360 - 200 = 160$

The total number of rasgulla boxes sold = $400 - 100 - 160 = 140$

Name of the shop	The total number of sweet boxes sold	The total number of laddu boxes sold	The total number of gulab jamun boxes sold	The total number of rasgulla boxes sold
Shop A	480	240	192	48
Shop B	720	216	360	144
Shop C	240	48	48	144
Shop D	400	160	100	140

Answer: C

The total number of laddu boxes sold in Shop B = 216

The total number of laddu boxes sold in Shop C = 48

The total number of gulab jamun boxes sold in Shop A = 192

The total number of gulab jamun boxes sold in Shop C = 48

Required difference = $(216 + 48) - (192 + 48) = 264 - 240 = 24$

Hence, the correct answer is option C

11. Questions

Answer: A

Let the monthly salary of Rajesh be $100x$

Money spent on groceries:

= 20% of monthly salary

= $(20 / 100) * 100x = 20x$

Remaining salary = $100x - 20x = 80x$

Money spent on buying books:

= 12.5% of $80x = 10x$

Money spent on rent:

= 7% of $80x = 5.6x$

Savings = $80x - 10x - 5.6x = 64.4x$

$5.6x = 5600$

$x = 1000$

$$\text{Monthly savings} = 64.4x = 64.4 * 1000$$

$$= \text{Rs. } 64400$$

$$\text{Annual savings} = 64400 * 12$$

$$= \text{Rs. } 772800$$

Hence, the correct answer is option A

12. Questions

Answer: D

Let the cost price of an article be $100x$

Marked price = 130% of cost price

$$= (130 / 100) * 100x = 130x$$

Selling price = 120% of cost price

$$= (120 / 100) * 100x = 120x$$

Marked price - Discount = Selling price

$$130x - 320 = 120x$$

$$130x - 120x = 320$$

$$10x = 320$$

$$x = 32$$

$$\text{Cost price of an article} = 100x = 100 * 32 = \text{Rs. } 3200$$

Required selling price = 145% of cost price

$$= (145 / 100) * 3200 = \text{Rs. } 4640$$

Hence, the correct answer is option D

13. Questions

Answer: D

Speed of the boat in still water : Speed of the stream = 13 : 5

Speed of the boat in still water = $13x$

Speed of the stream = $5x$

Given that,

Speed of the stream = 10 km/hr

$$5x = 10$$

$$x = 2$$

Speed of the boat in still water = $13x = 13 * 2 = 26 \text{ km/hr}$

Upstream speed = Speed of the boat in still water - Speed of the stream

$$\text{Upstream speed} = 26 - 10 = 16$$

Downstream speed = Speed of the boat in still water + Speed of the stream

$$\text{Downstream speed} = 26 + 10 = 36$$

According to the question,

$$((D - 30) / 16) + ((D + 40) / 36) = 3 (3/4)$$

$$(36D - 1080 + 16D + 640) = (15 / 4) * 16 * 36$$

$$52D - 440 = 2160$$

$$52D = 2160 + 440 = 2600$$

$$D = 2600 / 52 = 50$$

Hence, the correct answer is option D

14. Questions

Answer: A

According to the question,

$$\text{Total number of rings in a box (n)} = 5 + 10 + 3 = 18$$

The number of rings being picked (r) is 3

The total number of ways to select three rings

$${}^nC_r = n! / r!(n - r)!$$

$$\text{Total ways} = (18 * 17 * 16) / (3 * 2 * 1) = 816 \text{ ways}$$

$$\text{Ways to pick one gold ring} = 5 \text{ ways}$$

$$\text{Ways to pick one silver ring} = 10 \text{ ways}$$

$$\text{Ways to pick one diamond ring} = 3 \text{ ways}$$

$$\text{Favourable ways} = 5 * 10 * 3 = 150 \text{ ways}$$

$$\text{Required probability} = 150 / 816 = 25 / 136$$

The probability of getting one gold ring, one silver ring and one diamond ring is 25 / 136

Hence, the correct answer is option A

15. Questions

Answer: C

$$\text{S.I} = (p * n * r) / 100$$

Amount = Principle + Simple interest

$$7800 - p = \text{S.I}$$

$$7800 - p = (p * 8 * 7) / 100$$

$$7800 = ((56 / 100) * p) + p$$

$$7800 = (14 / 25) * p + p$$

$$7800 = p * ((14 / 25) + 1)$$

$$7800 = p * (39 / 25)$$

$$p = (7800 * 25) / 39$$

$$p = \text{Rs. } 5000$$

Compound interest:

$$p = \text{Rs. } 5000$$

$$\text{Amount} = p * (1 + (r / 100))^n$$

$$= 5000 * (1 + (10 / 100))^2$$

$$= 5000 * (110 / 100)^2$$

$$= 5000 * (121 / 100)$$

$$\text{Amount} = \text{Rs. } 6050$$

Hence, the correct answer is option C

16. Questions

Answer: D

Let the present age of Dad, Mom and son be a, b and c respectively

According to the question,

$$(a + b + c) / 3 = 45$$

$$(a + b + c) = 135 \rightarrow (1)$$

$$(a - 5) = b$$

$$a = (b + 5) \rightarrow (2)$$

$$(b - 10) / (c - 10) = 2 / 1$$

$$b - 10 = 2c - 20$$

$$2c - b = 10$$

$$2c = 10 + b$$

$$c = (10 + b) / 2 \rightarrow (3)$$

Substitute equation (2) and (3) in (1),

$$(b + 5) + b + ((10 + b) / 2) = 135$$

$$2b + 10 + 2b + 10 + b = 135 * 2$$

$$5b + 20 = 270$$

$$5b = 250$$

$$b = 250 / 5 = 50$$

The present age of Mom = 50 years

$$a = (b + 5)$$

$$a = 50 + 5 = 55$$

The present age of Dad = 55 years

$$\text{Required answer} = (55 + 5)$$

$$= 60 \text{ years}$$

Hence, the correct answer is option D

17. Questions

Answer: C

Let the initial quantity of juice and water be $4x$ and $5x$ respectively

After the addition of juice and water,

Juice increased by 25%:

$$= 125\% \text{ of } 4x = 5x$$

30 litres of water is added

$$= 5x + 30$$

According to the question,

$$(5x / (5x + 30)) = 1 / 2$$

$$10x = 5x + 30$$

$$5x = 30$$

$$x = 6$$

Initial quantity of juice = $4x = 4 * 6 = 24$ litres

Initial quantity of water = $5x = 5 * 6 = 30$ litres

Initial quantity of mixture in the vessel = $24 + 30 = 54$ litres

Hence, the correct answer is option C

18. Questions

Answer: E

If 12 men can complete 20% of the work in 5 days, then it takes 25 days to complete 100% of the work.

$$\text{Total work} = 12 * 25 = 300$$

One man's daily work rate = $(1 / 300)$ of the total work

If 15 women can complete 50% of the work in 10 days, then it takes 20 days to complete 100% of the work.

$$\text{Total work} = 15 * 20 = 300$$

One woman's daily work rate = $(1 / 300)$ of the total work

So for both a man and a woman, efficiency is equal.

$$\text{Combined rate of 5 men and 10 women} = (5 * (1 / 300)) + (10 * (1 / 300))$$

$$= (1 / 20) \text{ of the work per day}$$

$$\text{Required days} = 120\% \text{ of } 20$$

$$= (120 / 100) * 20 = 24 \text{ days}$$

Hence, the correct answer is option E

19. Questions

Answer: A

A's investment:

$$= (650 * x) + ((14 / 13) * 650) * (12 - x)$$

$$= (650 * x) + (700 * (12 - x))$$

$$= (650 * x) + 8400 - (700 * x)$$

$$= 8400 - (50 * x)$$

B's investment:

$$= (550 * x) + ((10 / 11) * 550) * (12 - x)$$

$$= (550 * x) + (500 * (12 - x))$$

$$= (550 * x) + 6000 - (500 * x)$$

$$= 6000 + (50 * x)$$

Profit share of A : Profit share of B = 5 : 4

$$(8400 - 50x) / (6000 + 50x) = 5 / 4$$

$$33600 - 200x = 30000 + 250x$$

$$33600 - 30000 = 450x$$

$$3600 = 450x$$

$$x = 3600 / 450$$

$$x = 8$$

Hence, the correct answer is option A

20. Questions

Answer: B

Perimeter of the circle = 88 cm

$$2\pi r = 88$$

$$2 * (22 / 7) * r = 88$$

$$r = (88 * 7) / (2 * 22)$$

$$r = 14 \text{ cm}$$

Since the area of a circle is the same as the base area of a cylinder, their radius must be equal.

Therefore, the radius of cylinder is 14 cm.

Curved surface area of cylinder = $2\pi rh$

$$2\pi rh = 1848$$

$$2 * (22 / 7) * 14 * h = 1848$$

$$(616 / 7) * h = 1848$$

$$h = (1848 * 7) / 616$$

$$h = 21 \text{ cm}$$

Hence, the correct answer is option B

21. Questions

Answer: A

$$98\% \text{ of } 300 + (720 / 12) = (?) * 3$$

$$(98 / 100) * 300 + (720 / 12) = (?) * 3$$

$$294 + 60 = (?) * 3$$

$$354 = (?) * 3$$

$$(?) = 118$$

Hence, the correct answer is option A

22. Questions

Answer: D

$$\sqrt{841} + (145 / 5) - (8 * 6) = (?)$$

$$29 + 29 - 48 = (?)$$

$$58 - 48 = (?)$$

$$(?) = 10$$

Hence, the correct answer is option D

23. Questions

Answer: C

$$(12 * 16 * 32) / (8 * 2 * 6) = \sqrt{(?)} * 4$$

$$6144 / 96 = \sqrt{(?)} * 4$$

$$64 = \sqrt{(?)} * 4$$

$$\sqrt{(?)} = 64 / 4$$

$$(?) = 16^2$$

$$(?) = 256$$

Hence, the correct answer is option C

24. Questions

Answer: B

$$(98 / 2) - (143 / 11) = (?) - \sqrt{49} + 19$$

$$49 - 13 = (?) - 7 + 19$$

$$36 + 7 - 19 = (?)$$

$$43 - 19 = (?)$$

$$(?) = 24$$

Hence, the correct answer is option B

25. Questions

Answer: E

$$(12 * 15) + (7 * 6) - (9 * 3) = (?) + 35$$

$$180 + 42 - 27 - 35 = (?)$$

$$222 - 62 = (?)$$

$$160 = (?)$$

Hence, the correct answer is option E

26. Questions

Answer: B

$$23.90 + (123.88 / 4.09) = (?)^2 - (13.15 * 1.89)$$

$$24 + (124 / 4) = (?)^2 - (13 * 2)$$

$$24 + 31 = (?)^2 - 26$$

$$55 + 26 = (?)^2$$

$$(?)^2 = 81$$

$$(?) = \sqrt{81} = 9$$

Hence, the correct answer is option B

27. Questions

Answer: A

$$45.12\% \text{ of } 579.91 - (155.94 / 13.04) = (?) * 3.01$$

$$45\% \text{ of } 580 - (156 / 13) = (?) * 3$$

$$(45 / 100) * 580 - 12 = (?) * 3$$

$$261 - 12 = (?) * 3$$

$$249 = (?) * 3$$

$$(?) = 249 / 3 = 83$$

Hence, the correct answer is option A

28. Questions

Answer: D

$$\sqrt{(12.90 + 23.13)} = (?) - (14.11 * 14.89) + (2.07)^4$$

$$\sqrt{(13 + 23)} = (?) - (14 * 15) + (2)^4$$

$$\sqrt{36} = (?) - 210 + 16$$

$$6 = (?) - 194$$

$$(?) = 194 + 6 = 200$$

Hence, the correct answer is option D

29. Questions

Answer: B

$$(5 * 9) - \sqrt{36} + (11 * 12) = (?)$$

$$45 - 6 + 132 = (?)$$

$$177 - 6 = (?)$$

$$(?) = 171$$

Hence, the correct answer is option B

30. Questions

Answer: D

$$(189.19 / 8.88) + \sqrt{(91.21 - 65.92)} = (?)^3 - 38.12$$

$$(189 / 9) + \sqrt{(91 - 66)} = (?)^3 - 38$$

$$21 + \sqrt{25} + 38 = (?)^3$$

$$21 + 5 + 38 = (?)^3$$

$$(?)^3 = 64$$

$$(?) = \sqrt[3]{64}$$

$$(?) = 4$$

Hence, the correct answer is option D

31. Questions

Answer: B

The cube of a prime number increased by 1

$$(2)^3 + 1 = 9$$

$$(3)^3 + 1 = 28$$

$$(5)^3 + 1 = 126$$

$$(7)^3 + 1 = \mathbf{344}$$

$$(11)^3 + 1 = 1332$$

$$(13)^3 + 1 = 2198$$

Hence, the correct answer is option B

32. Questions

Answer: A

123	201	292	398	521	663	826
78	91	106	123	142	163	
+13	+15	+17	+19	+21		

Hence, the correct answer is option A

33. Questions

Answer: B

the number being added doubles at each step.

$$71 + 6 = 77$$

$$77 + 12 = 89$$

$$89 + 24 = \mathbf{113}$$

$$113 + 48 = 161$$

$$161 + 96 = 257$$

$$257 + 192 = 449$$

Hence, the correct answer is option B

34. Questions

Answer: E

add the squares of consecutive integers, starting from 11^2

$$35.5 + (11)^2 = 156.5$$

$$156.5 + (12)^2 = 300.5$$

$$300.5 + (13)^2 = 469.5$$

$$469.5 + (14)^2 = 665.5$$

$$665.5 + (15)^2 = \mathbf{890.5}$$

$$890.5 + (16)^2 = 1146.5$$

Hence, the correct answer is option E

35. Questions

Answer: B

multiplied and divided by consecutive integers starting from 2

$$26.25 * 2 = 52.5$$

$$52.5 / 3 = \mathbf{17.5}$$

$$17.5 * 4 = 70$$

$$70 / 5 = 14$$

$$14 * 6 = 84$$

$$84 / 7 = 12$$

Hence, the correct answer is option B

36. Questions

Answer: E

Subtracting the numbers by prime numbers starting 41

$$259 - 41 = \mathbf{218}$$

$$218 - 43 = 175$$

$$175 - 47 = 128$$

$$128 - 53 = 75$$

$$75 - 59 = 16$$

Hence, the correct answer is option E

37. Questions

Answer: B

Continuous division of the number, where the divisor decreases by exactly 0.5 at each step

$$7560 / 4.5 = 1680$$

$$1680 / 4 = 420$$

$$420 / 3.5 = 120$$

$$120 / 3 = \mathbf{40}$$

$$40 / 2.5 = 16$$

$$16 / 2 = 8$$

Hence, the correct answer is option B

38. Questions

Answer: C

adding an increasing integer inside the brackets, then multiply the result by that same increasing integer

$$(11 + 1) * 2 = 24$$

$$(24 + 2) * 3 = \mathbf{78}$$

$$(78 + 3) * 4 = 324$$

$$(324 + 4) * 5 = 1640$$

$$(1640 + 5) * 6 = 9870$$

Hence, the correct answer is option C

39. Questions

Answer: C

adding consecutive multiples of 8 at each step

$$134 + (8 * 1) = 142$$

$$142 + (8 * 2) = 158$$

$$158 + (8 * 3) = 182$$

$$182 + (8 * 4) = 214$$

$$214 + (8 * 5) = \mathbf{254}$$

$$254 + (8 * 6) = 302$$

Hence, the correct answer is option C

40. Questions

Answer: C

alternates between subtraction and addition using the squares of consecutive integers, starting from 4^2

$$186 - (4)^2 = 170$$

$$170 + (5)^2 = 195$$

$$195 - (6)^2 = \mathbf{159}$$

$$159 + (7)^2 = 208$$

$$208 - (8)^2 = 144$$

$$144 + (9)^2 = 225$$

Hence, the correct answer is option C

41. Questions

Answer: A

I). $3x^2 - 41x + 136 = 0$

$$3x^2 - 24x - 17x + 136 = 0$$

$$3x(x - 8) - 17(x - 8) = 0$$

$$(3x - 17)(x - 8) = 0$$

$$x = (17 / 3), 8$$

$$x = 5.66, 8$$

II). $4y^2 - 68y + 288 = 0$

$$4y^2 - 32y - 36y + 288 = 0$$

$$4y(y - 8) - 36(y - 8) = 0$$

$$(4y - 36)(y - 8) = 0$$

$$y = (36 / 4), 8$$

$$y = 9, 8$$

Value of x	Value of y	Comparison
5.66	9	$x < y$
8	9	$x < y$
5.66	8	$x < y$
8	8	$x = y$

Hence, $x \leq y$

Hence, the correct answer is option A

42. Questions

Answer: B

$$4x + y = 15$$

$$y = 15 - 4x \rightarrow (1)$$

$$5x + 6y = 14 \rightarrow (2)$$

Substitute equation (1) in (2),

$$5x + 6 * (15 - 4x) = 14$$

$$5x + 90 - 24x = 14$$

$$90 - 14 = 24x - 5x$$

$$19x = 76$$

$$x = 76 / 19 = 4$$

Substitute $x = 4$ in equation (1),

$$y = 15 - (4 * 4)$$

$$y = 15 - 16 = -1$$

Value of x	Value of y	Comparison
4	-1	$x > y$

Hence, $x > y$

Hence, the correct answer is option B

43. Questions

Answer: D

I). $x^2 - 6x - 135 = 0$

$$x^2 - 15x + 9x - 135 = 0$$

$$x(x - 15) + 9(x - 15) = 0$$

$$(x + 9)(x - 15) = 0$$

$$x = -9, 15$$

II). $y^2 - 32y + 87 = 0$

$$y^2 - 29y - 3y + 87 = 0$$

$$y(y - 29) - 3(y - 29) = 0$$

$$(y - 29)(y - 3) = 0$$

$$y = 29, 3$$

Value of x	Value of y	Comparison
-9	29	$x < y$
15	29	$x < y$
-9	3	$x < y$
15	3	$x > y$

Hence, $x = y$ or relationship cannot be established

Hence, the correct answer is option D

44. Questions

Answer: A

I). $x^2 + 15x - 450 = 0$

$$x^2 - 30x + 15x - 450 = 0$$

$$x(x - 30) + 15(x - 30) = 0$$

$$(x + 15)(x - 30) = 0$$

$$x = -15, 30$$

II). $y^2 + 34y + 288 = 0$

$$y^2 + 16y + 18y + 288 = 0$$

$$y(y + 16) + 18(y + 16) = 0$$

$$(y + 16)(y + 18) = 0$$

$$y = -16, -18$$

Value of x	Value of y	Comparison
-15	-16	$x > y$
30	-16	$x > y$
-15	-18	$x > y$
30	-18	$x > y$

Hence, $x > y$

Hence, the correct answer is option A

45. Questions

Answer: E

I). $x^2 = \sqrt{49} + 29$

$$x^2 = 7 + 29 = 36$$

$$x = \sqrt{36}$$

$$x = +6, -6$$

II). $y^2 - 17y + 72 = 0$

$$y^2 - 8y - 9y + 72 = 0$$

$$y(y - 8) - 9(y - 8) = 0$$

$$(y - 8)(y - 9) = 0$$

$$y = 8, 9$$

Value of x	Value of y	Comparison
-6	8	$x < y$
6	8	$x < y$
-6	9	$x < y$
6	9	$x < y$

Hence, $x < y$

Hence, the correct answer is option E

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